

EXPERIMENTAL DETERMINATION OF Z PARAMETERS

AIM :

TO calculate and verify z parameters of a two port network using Multisim Simulator.

APPARATUS REQUIRED:

S.NO	Components	Specification	Quantity
1.	Power Supply		1
2.	Multimeter		1
3.	Resistors	$10k\Omega, 10k\Omega, 4.7k\Omega$	
4.	Bread Board		1
5.	connecting wires		
6.	PC with Multisim simulator		

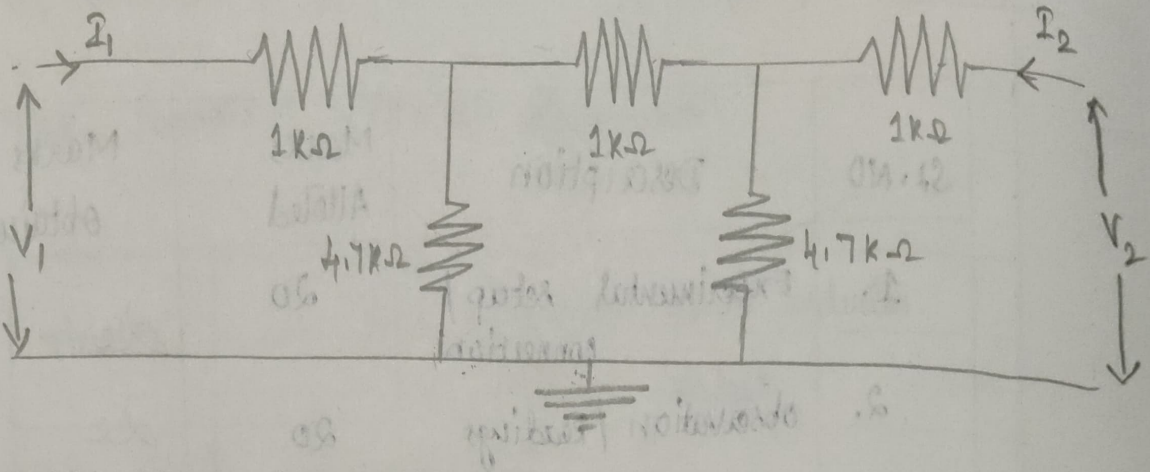
PROCEDURE :

a) connect the circuit as shown in Fig. & switch 'ON' the experimental board.

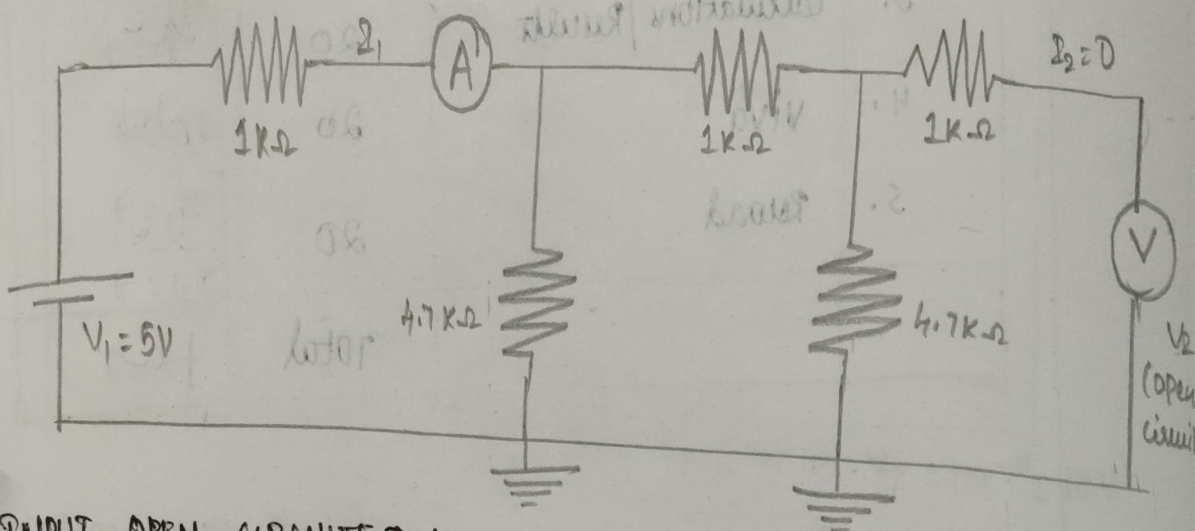
b) First open the o/p terminal & supply 5V to I/P terminal. Measure o/p voltage and I/P current.

c) Secondly, open I/P terminal & supply 5V to o/p terminal. Measure I/P voltage & o/p current.

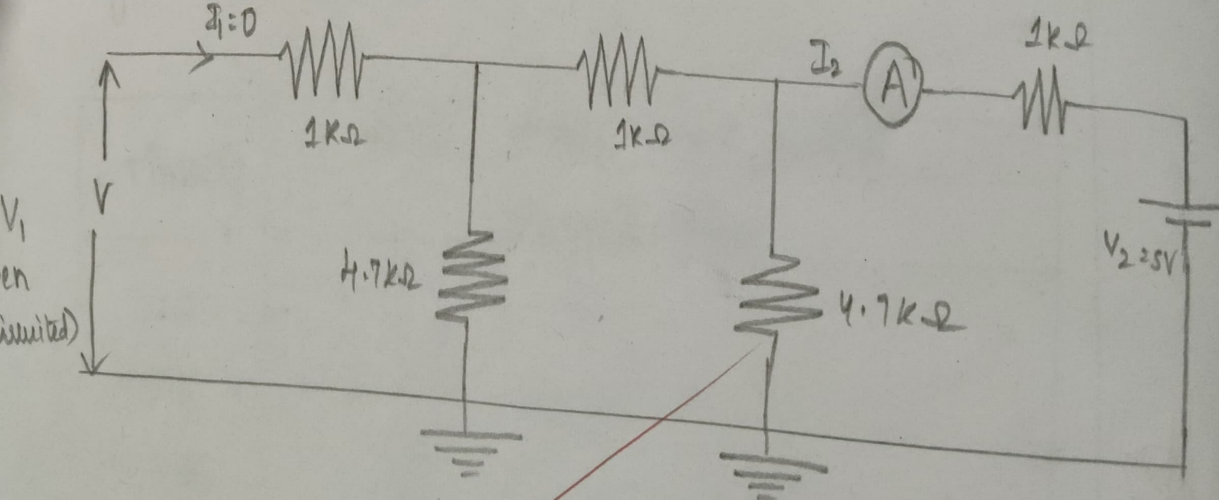
CIRCUIT DIAGRAM:



OUTPUT OPEN CIRCUITED:



INPUT OPEN CIRCUITED:



OBSERVATION :

S.NO	OUTPUT OPEN CIRCUITED ($I_2=0$)			$Z_{21} = V_2 / I_1 (\Omega)$	$Z_{11} = V_1 / I_1 (\Omega)$
	V_1	V_2	I_1		
1.	5.0	2.60	2.10	1.23 k Ω	2.38 k Ω
2.	5.0	2.55	2.08	1.22 k Ω	2.40 k Ω
3.	5.0	2.63	2.12	1.24 k Ω	2.35 k Ω

S.NO	INPUT OPEN CIRCUITED ($I_1=0$)			$Z_{12} = V_1 / I_2 (\Omega)$	$Z_{22} = V_2 / I_2 (\Omega)$
	V_2	V_1	I_2		
1.	5.0	2.60	2.10	1.23	4.00
2.	5.0	2.57	2.09	1.22	2.39
3.	5.0	2.62	2.12	1.24	2.38

- d) Calculate the values of Z parameters using equation (1) & (2).
- e) Switch 'OFF' the supply after taking the readings.

MARKS SPLIT-UP:

Sl. NO	Description	Marks allotted	Marks Obtained
1.	Experimental setup	20	20
2.	Observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20
		Total	100

RESULT:

The Z -parameters of the two port network has been calculated & verified by using Multisim simulator.

